

# Face-Rotation BFS Nets of the Six Regular Convex 4-Polytopes

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# Unfolding a 4-polytope

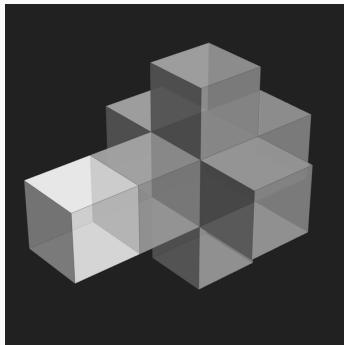
A **net** of a convex polytope: cut it open and lay the facets out flat, without overlap.

In 3D this is Dürer's problem, still open.

In 4D the facets are **3-dimensional cells** and the net lives in  $\mathbb{R}^3$ .

- Turney (1984): all 261 nets of the 8-cell
- Devadoss–Harvey (2023): *every* spanning tree works for the 5-, 8-, 16-cell

**Left open: 24-cell, 120-cell, 600-cell.**



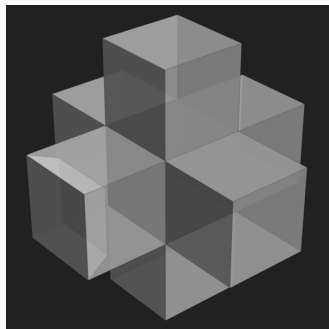
A net of the 8-cell: Dalí's cross

# Face-rotation BFS

Fix a **root cell**  $r$ . Three steps.

- 1 **Face-up.** Rotate in  $SO(4)$  so the centroid of  $r$  sits on the  $+w$  axis.
- 2 **BFS.** Take a breadth-first spanning tree of the cell-adjacency graph from  $r$ . For each tree edge (parent  $p$ , child  $c$ ), rotate  $c$  about the shared polygon until it lands in  $p$ 's hyperplane.
- 3 **Project.** All cells now share one  $w$ -coordinate (spread  $< 4 \times 10^{-13}$ ). Drop  $w$ .

The result is a net iff no two cells share interior volume.



Opening one BFS layer at a time, into Dalí's cross

## Step 2 hides a choice

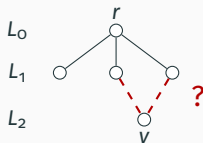
The layers are forced:  $L_d$  is the set of cells at distance  $d$  from  $r$ , and no rule can change that.

The **parent** is not. A cell in  $L_d$  may touch several cells of  $L_{d-1}$ , and BFS takes whichever one the implementation happens to reach first.

### A geometric rule need not help

For the 5-, 8-, 16- and 24-cell the symmetries fixing  $r$  and  $v$  permute the candidates transitively, so **every** equivariant score — height along  $w$ , turning angle, the volume form — ties on all of them.

Ours breaks the tie by cell index, which is not equivariant either. **We come back to this.**



Two admissible parents  
for  $v$

## The validity test, and a trap

A net is valid if for every pair of cells not joined by a tree edge,

$$\text{Vol}_3(\text{conv}(\mathbf{c}_i) \cap \text{conv}(\mathbf{c}_j)) \leq 10^{-8}.$$

### Boundary contact is not overlap

Testing with closed-region membership instead counts a shared face, edge or vertex as an intersection. That is common for pointed cells, and it wrongly reports

16-cell 0/16,    24-cell 0/24,    600-cell 0/600.

Measuring *volume* ignores contact of measure zero.

# Main result

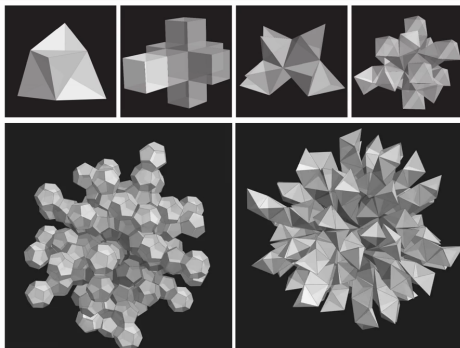
## Theorem

For every regular convex 4-polytope and **every** root cell, the face-rotation BFS net is valid.

$5 + 8 + 16 + 24 + 120 + 600 = 773$   
roots, all verified.

New beyond

Devadoss–Harvey: the 24-,  
120- and 600-cell.



## Contrast: the apple-peel algorithm

A different family of spanning trees — a greedy spiral/zonal ordering (Yoshino–Chaidee, [arXiv:2605.30373](#)).

| Polytope | BFS<br>valid | apple-peel (RZ) |        |
|----------|--------------|-----------------|--------|
|          |              | orderings       | nets   |
| 120-cell | all roots    | 1440/1440       | 0/1440 |
| 600-cell | all roots    | 0/2400          | —      |

The 120-cell has an ordering for every start and **never** a net; the 600-cell has no ordering at all.

### Consequence

Having a valid *ordering* and having a valid *net* are independent properties.

## And more: the tie-breaking question

The abstract left this open:

*“Whether validity extends to all BFS tie-breakings ... remains open.”*

BFS layers do not depend on the rule, so a BFS tree is just a choice of parent, one layer up, for each cell. They can then be counted with no rule at all.

|          | trees / root              | checked   |
|----------|---------------------------|-----------|
| 5-cell   | 1                         | all       |
| 8-cell   | 6                         | all       |
| 16-cell  | 20,736                    | all       |
| 24-cell  | 32,768                    | all       |
| 120-cell | $\sim 1.8 \times 10^{44}$ | 20 random |
| 600-cell | $\sim 1.8 \times 10^{65}$ | 20 random |

**1,132,661 trees, no self-intersection.** For the first four the rule is irrelevant. And BFS is not a vacuous restriction: the 600-cell (Devadoss–Harvey) and the 24-cell (Seamons 2026) are **not all-net**.

## △ Warning: this does not extend

Beyond the regular case the rule stops being harmless.

### Grand antiprism, 320 cells

rule A    valid from **all 320** roots

rule B    valid from **no** root

Same polytope, same roots, same family of BFS trees. Only the choice of parent differs.

Among the 64 convex uniform 4-polytopes this is not isolated: another swings from 1.8% to 99.8% of its roots.

**What regularity buys is not that some tree works, but that the choice does not matter.**



Grand antiprism, one valid net

## Open problems

- Why? A proof that does not enumerate.
- From BFS trees to **all** spanning trees: the 600-cell is not all-net (Devadoss–Harvey), nor is the 24-cell (Seamons 2026).  
**Only the 120-cell is left.**
- For the 120- and 600-cell even the BFS family is open:  $10^{44}$  and  $10^{65}$  trees per root put enumeration out of reach, so an argument would have to be structural.
- What separates the uniform polytopes that are insensitive to the rule from those that are not?

Code, data and nets:

`github.com/takashi-randomwalker/apple-peel-4d`

## Backup: “BFS tree” is bigger than what BFS returns

A traversal reaches a tree iff its parent choices admit a total order, i.e. iff the constraint digraph  $p(v) \rightarrow q$  is acyclic.

|         | BFS trees | reachable |       |
|---------|-----------|-----------|-------|
| 8-cell  | 6         | 6         | 100%  |
| 16-cell | 20,736    | 4,896     | 23.6% |
| 24-cell | 32,768    | 2,592     | 7.9%  |

So of the 786,432 trees verified for the 24-cell, only 62,208 are reachable by any traversal. The obstruction is a *cycle* in the constraints, not merely two cells sharing two candidate parents: the 16-cell has no such pair and still leaves three quarters of its trees unreachable.

## Backup: which rules were compared

|                         | queue     | max-index | min-index |
|-------------------------|-----------|-----------|-----------|
| Runcinated 24-cell      | 0/240     | 0/240     | 0/240     |
| Runcitruncated 600-cell | 47/2640   | 2635/2640 | 2627/2640 |
| Runcitruncated 120-cell | 2570/2640 | 2640/2640 | 2151/2640 |
| Grand antiprism         | 320/320   | 320/320   | 0/320     |

“queue” is the textbook BFS loop, neighbours scanned in index order.  
“min/max-index” choose, for each cell, the smallest or largest indexed candidate parent.

Uniformly random trees are harsher still: of the 61 uniform polytopes valid from every root under the queue rule, 13 fail within 10 random trees, the grand antiprism among them — and every one of the four polytopes with 2,640 cells fails all ten, whether or not it is one of the 61.

## Backup: the one that no rule could unfold

Runcinated 24-cell  $t_{0,3}\{3, 4, 3\}$ , 240 cells, 0/240 under all three index rules. Ask instead for a *symmetric tree*: a BFS tree invariant under  $H \leq \text{Stab}(r)$  exists iff every cell has a candidate parent fixed by its stabilizer in  $H$ . Here  $|\text{Aut}| = 2,304$ , with two cell orbits, and the largest such  $H$  has order 3.

| root orbit              | $ \text{Stab}(r) $ | $\max  H $ | $H$ -invariant | random |
|-------------------------|--------------------|------------|----------------|--------|
| 48 (octahedra)          | 48                 | 3          | 70/600         | 0/600  |
| 192 (triangular prisms) | 12                 | 3          | 83/600         | 0/600  |

The control is the point: unconstrained trees at those roots give nothing. Transitivity on each orbit then gives a valid net from **all 240 roots** — not a polytope without a net, but one whose nets no index rule finds.

(No conflict with slide 4: there the candidates form one orbit under the *full* stabilizer, so no invariant tree exists at all.)

## Backup: what does *not* separate them

**Cell degree** fails both ways. The grand antiprism and the snub 24-cell have cells of degree 4 — the poverty that stops apple-peel on the 600-cell — yet are valid from every root, while the runcinated 24-cell has minimum degree 5 and is valid from none. Nor does high degree help: every prism has a cell of degree up to 92 and all 17 are fine, and the root-dependent runcitruncated 120-cell carries cells of degree 32.

**Angular defect at an edge** —  $2\pi$  less the dihedral angles meeting there, zero exactly when those cells tile  $\mathbb{R}^3$  around it. For the regular six it is one number, and it orders them *exactly* by how hard they proved to be:  $148.4^\circ$ ,  $90^\circ$ ,  $77.9^\circ$ ,  $31.6^\circ$ ,  $10.3^\circ$ ,  $7.4^\circ$ . Among the uniform ones it breaks both ways: the omnitruncated 120-cell shares its minimum defect,  $0.736^\circ$ , with the root-dependent runcitruncated 120-cell and is valid from every root, while the runcinated 24-cell sits at  $10.5^\circ$ , above many that are. Weighting by depth or by size does not repair it.

## Backup: the six polytopes

| Polytope | Symbol        | $V$ | $F$  | $C$ | Cell         | deg | depth | trees / root         |
|----------|---------------|-----|------|-----|--------------|-----|-------|----------------------|
| 5-cell   | $\{3, 3, 3\}$ | 5   | 10   | 5   | tetrahedron  | 4   | 1     | 1                    |
| 8-cell   | $\{4, 3, 3\}$ | 16  | 24   | 8   | cube         | 6   | 2     | 6                    |
| 16-cell  | $\{3, 3, 4\}$ | 8   | 32   | 16  | tetrahedron  | 4   | 4     | 20,736               |
| 24-cell  | $\{3, 4, 3\}$ | 24  | 96   | 24  | octahedron   | 8   | 3     | 32,768               |
| 120-cell | $\{5, 3, 3\}$ | 600 | 720  | 120 | dodecahedron | 12  | 5     | $1.8 \times 10^{44}$ |
| 600-cell | $\{3, 3, 5\}$ | 120 | 1200 | 600 | tetrahedron  | 4   | 15    | $1.8 \times 10^{65}$ |

*deg* counts the cells sharing a 2-face; *depth* is the eccentricity of the cell-adjacency graph, the same from every cell. The layers are 1+4; 1+6+1; 1+4+6+4+1; 1+8+14+1; 1+12+32+42+32+1; and sixteen layers for the 600-cell.

The 24-cell row tells the two families apart: every BFS tree of it fits in 1+8+14+1, whereas the overlapping spanning tree of Seamon's puts 13 of its 24 cells deeper than their distance from the root.